

COMPACT CURVE-FLAT QUOTIENT REALIZATION: LITERATURE STATUS

AGENT LANE 10

STATUS

Status: `literature_already_answered`.

SOURCE QUESTION

In Remark 6.9 of Flores–Jung–Lancien–Petitjean–Prochazka–Quilis, *On curve-flat Lipschitz functions and their linearizations* [1], the authors give a compact example whose curve-flat quotient is the interval $[0, 1]$. They then propose the following compact realization problem: study for which non-purely 1-unrectifiable compact metric spaces K there exists a compact metric space M such that

$$M_{\text{CF}} = K.$$

SUPPORTING ANSWER

Kaasik–Quilis [2] explicitly cite this question in their introduction. They explain that their exact complete-space theorem does not settle the compact version because the constructed space is not compact unless the target is finite, and then state that they modify the construction in order to solve the compact problem, with the caveat that exact isometry is not always obtained.

The answer supplied by [2] is therefore scoped as follows. Their Theorem B states that for every compact metric space (M, d) , every countable ordinal α , and every $\varepsilon > 0$, there is a compact metric space (N, ρ) with $\alpha_{cf}(N) \geq \alpha$ such that

$$N/\rho_{cf}^\alpha$$

is $(1+\varepsilon)$ -isometric to (M, d) . The more technical Theorem 3.1 gives exact isometry in particular when the compact target is finite, geodesic, or a gapped segment.

SCOPE LIMITATIONS

This is not recorded as a new proof by the run. It is a separate later-paper answer to the source paper’s compact curve-flat quotient realization program. The all-compact statement is approximate up to arbitrarily small bi-Lipschitz distortion. The packet does not assert that every compact metric space has an exact compact first-order curve-flat preimage.

CITATION EVIDENCE

The supporting paper explicitly names the question from [1]: it says that Theorem A does not answer that question because the constructed space need not be compact, and then introduces the compact modification leading to Theorem B. Thus this belongs in the literature-already-answered folder, not in a new full or partial proof folder.

REFERENCES

- [1] G. Flores, M. Jung, G. Lancien, C. Petitjean, A. Prochazka, and A. Quilis, *On curve-flat Lipschitz functions and their linearizations*, arXiv:2411.08369.
- [2] J. K. Kaasik and A. Quilis, *Curve-flat functions and Lipschitz quotients*, arXiv:2603.20177.